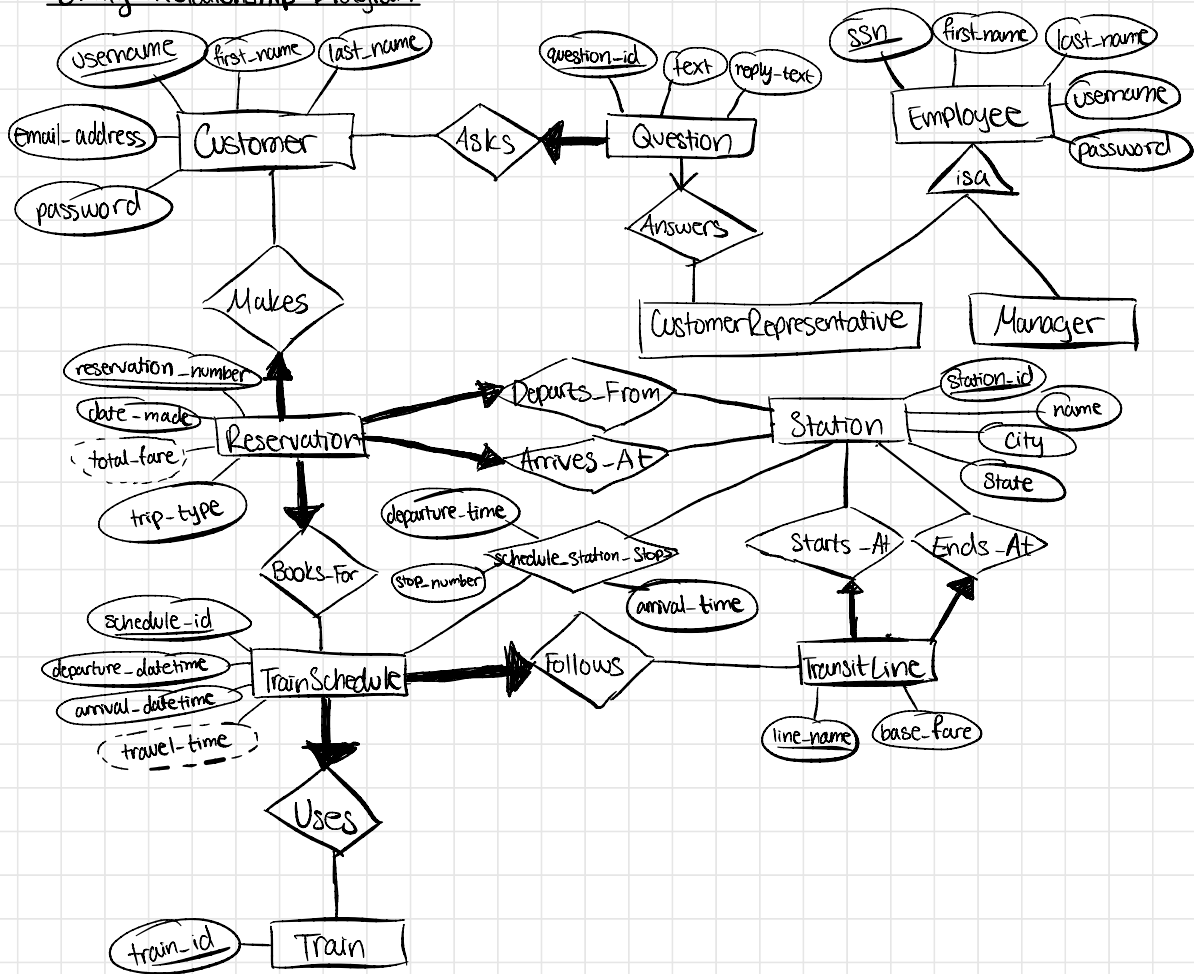


Entity Relationship DiagramConstraints

- ① Fare calculation rules (user demographics / round trip)
- ② Manager and customer representative permissions

Relational Schema

Customer (username, first-name, last-name, email-address, password,
primary key (username))

Employee (ssn, first-name, last-name, username, password, role,
primary key (ssn))

Train (train-id,
primary key (train-id))

Station (station-id, name, city, state,
primary key (station-id))

TransitLine (line-name, base-fare, starts-station-id, ends-station-id,
primary key (line-name),
foreign key (starts-station-id) references Station(station-id),
foreign key (ends-station-id) references Station(station-id))

TrainSchedule (schedule-id, departure-datetime, arrival-datetime, line-name,
train-id,
primary key (schedule-id),
foreign key (line-name) references TransitLine (line-name) not null,
foreign key (train-id) references Train (train-id) not null)

Schedule-Station-Stops (schedule-id, station-id, departure-time, arrival-time, stop-number,
primary key (schedule-id, station-id),
foreign key (schedule-id) references TrainSchedule (schedule-id),
foreign key (station-id) references Station (station-id))

Reservation (reservation-number, date-made, total-fare, trip-type,
customer-username, schedule-id, departure-station-id,
arrival-station-id,
primary key (reservation-number),
foreign key (customer-username) references Customer (username) not null,
foreign key (schedule-id) references TrainSchedule (schedule-id) not null,
foreign key (departure-station-id) references Station (station-id) not null,
foreign key (arrival-station-id) references Station (station-id) not null)

Question (question-id, text, reply-text, customer-username, rep-ssn,
primary key (question-id),
foreign key (customer-username) references Customer (username) not null,
foreign key (rep-ssn) references Employee (ssn))